MVRL LAB MEETING · 23 OCT 2026

Does Remote Sensing Beat the Trail Map?

Predicting hiking speed in regions a model has never seen

Brian Zhou
CSE 5999 independent study · Mentors Danni Beaulieu and Tianyi Gao · Advisor Prof. Nathan Jacobs

Background: 10 m contours of Tuckerman and Huntington ravines, Mt Washington, NH, from USGS 3DEP 1 m lidar. Slides prepared with help from Claude (Anthropic).

GIS still times hikes with a 1993 formula

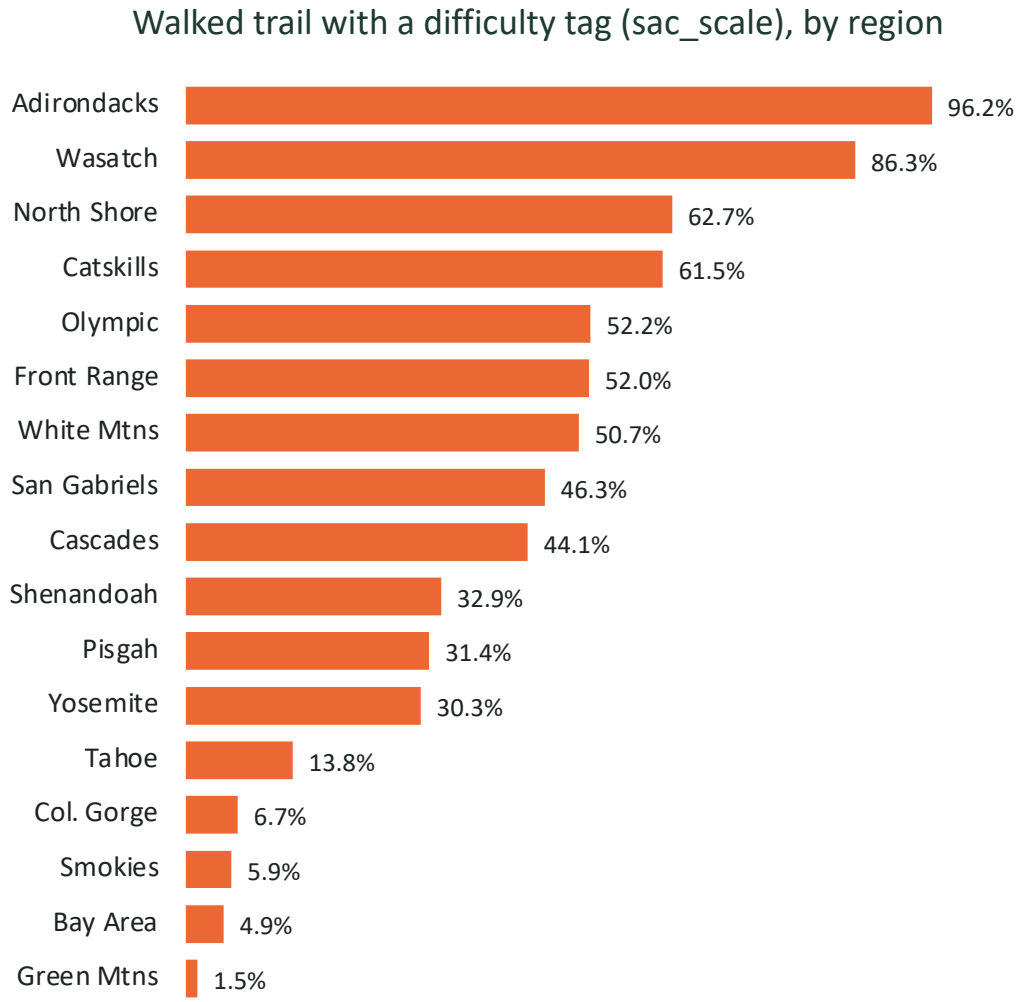
TOBLER'S HIKING FUNCTION (1993)

$$v = 6\, e^{-3.5\,|s + 0.05|} \text{ km/h}$$

s = slope (0.10 = 10% uphill). Fastest at a gentle downhill: 6 km/h. Flat: 5.0 km/h. 20% climb: 2.5 km/h.

- **Fit to figures in a Swiss cartography textbook**
  not to observed hikes
- **One curve for every trail**
  no rocks, roots, steps or vegetation
- **Still the default in GIS travel-time tools**
  search and rescue, wildfire escape routes, trail apps

And the map rarely helps: OSM's difficulty tag covers **1.5% to 96%** of walked trail, depending on the region.



Prior work stays inside one region

Slope-only curves
Tobler 1993; Campbell et al. 2019, 2022 (Strava, AllTrails): Tobler is too fast

Extra inputs, one study area
Wood et al. 2023 (England & Wales, OSM + lidar); Huang et al. 2026 (Hong Kong): random or cross-validated splits

Lidar model from field walks
STRIDE, Campbell et al. 2024: slope, understory density, ground roughness

This lab
RasterNet, Hadzic et al. 2020: road speed from lidar and overhead imagery

Missing: a test in regions the model never saw, and map tags vs remote sensing head to head.

OUR GOAL

Predict the speed of every 50 m of trail, with its uncertainty, from what can be seen from above, and test it in regions the model never saw.

Four models, same method, different inputs

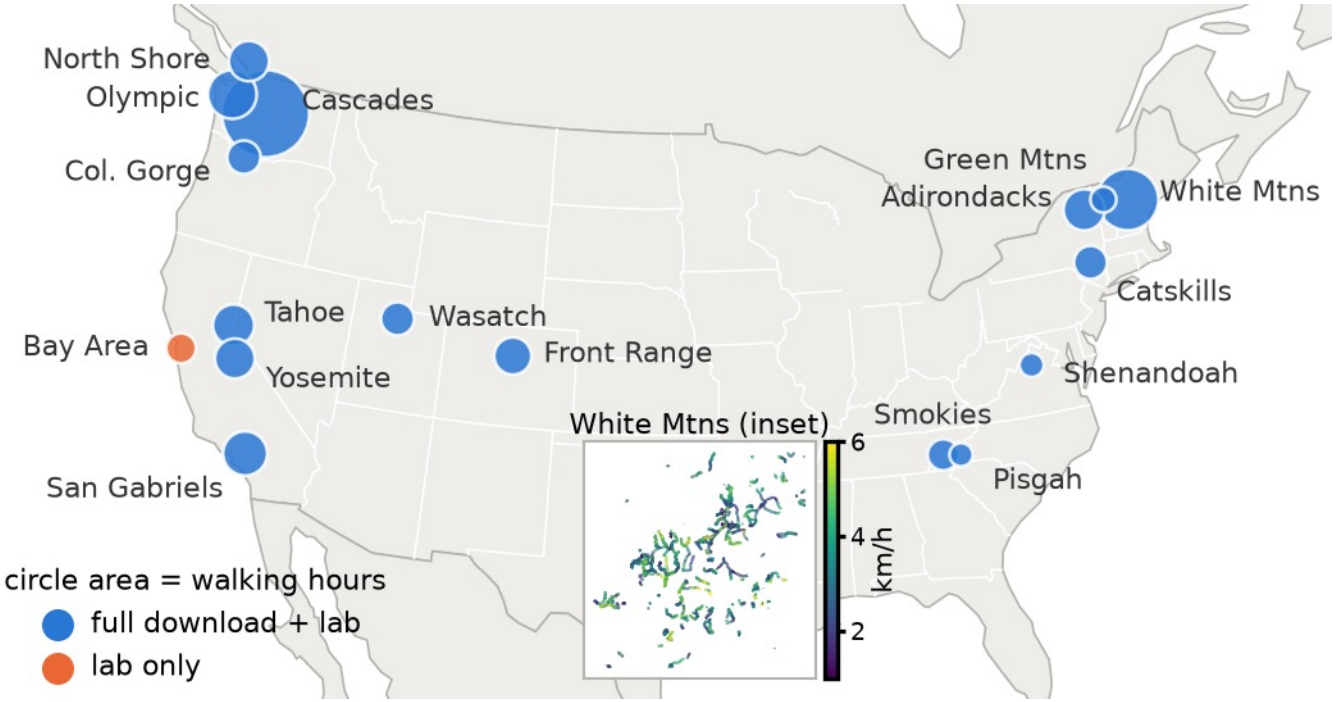
Tobler
slope only

Tags
OSM trail tags

Remote sensing
lidar terrain, vegetation, embeddings

Both
everything

17 regions, 290,640 measured trail segments



Circle area = walking hours. Blue: full OSM download plus the lab's crawl; orange: crawl only. Inset: White Mountains segments by speed.

290,640

walking segments, 50 m each

1,992

hikes (outings)

4,702 h

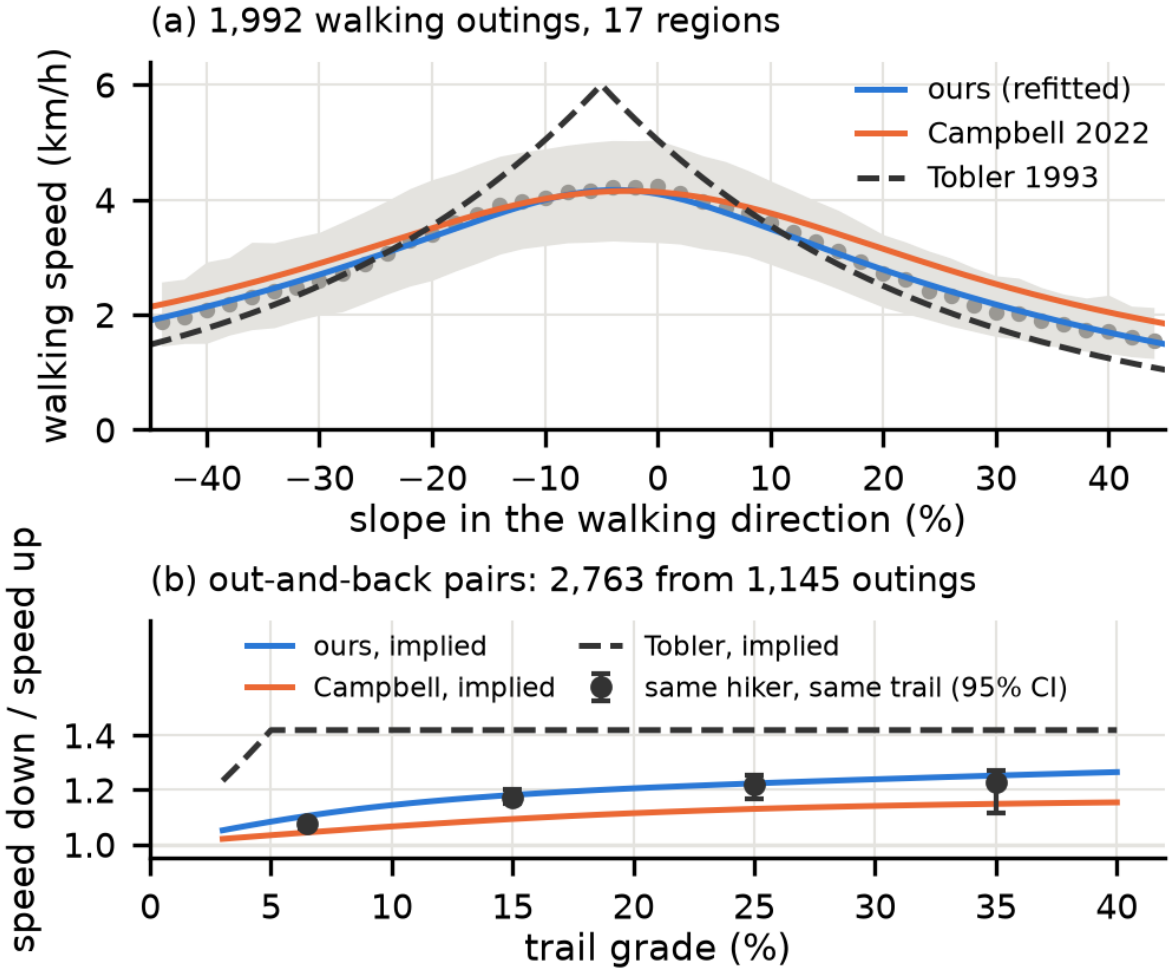
of walking

14,532 km

of trail walked

Public OSM GPS traces → Clean and remove stops → Match to the trail map → Cut into 50 m segments → **Add slope, tags, lidar terrain, vegetation, embeddings**

Tobler is too fast and too peaked



4.2 km/h

real peak walking speed, on level to gently descending trail

Tobler: 6 km/h

8–23%

faster downhill than uphill, same hiker on the same trail

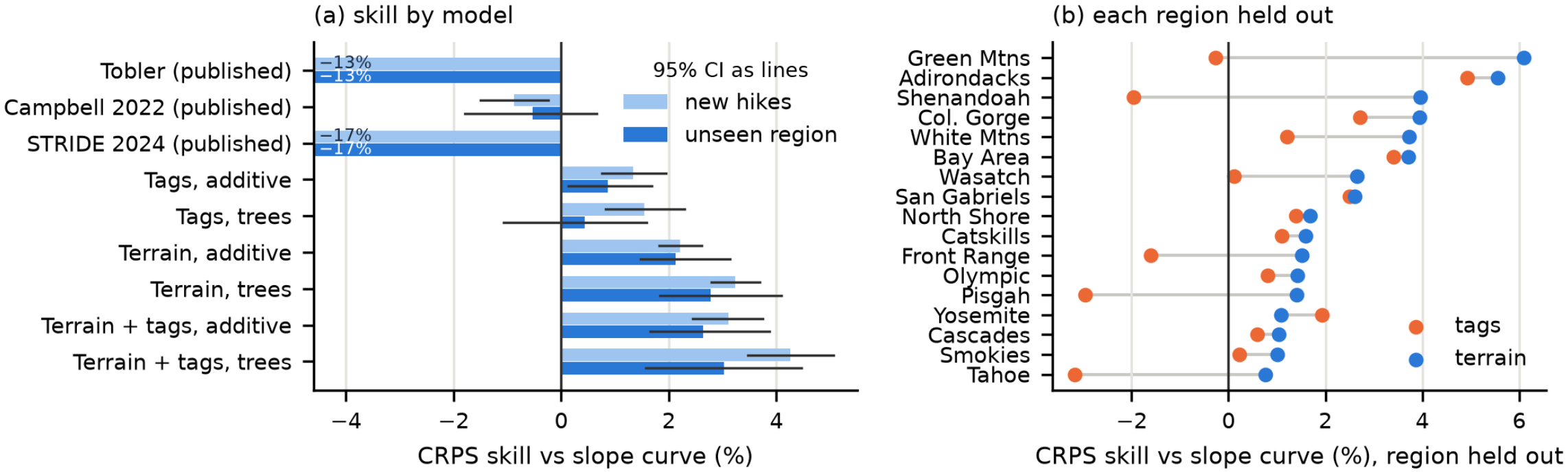
Tobler: 42% at every grade

17 of 17

regions where Tobler predicts worse than a refitted slope curve

–12.5% overall (CRPS skill)

In new regions, terrain beats map tags



+2.1 to +2.8%

terrain, in regions the model never saw

16 of 17

held-out regions where terrain beats tags

+0.4 to +0.9%

OSM tags, in regions the model never saw

Skill: how much better a model's predicted speed distribution scores (CRPS) than a slope-only curve refitted to our data. It holds for newer hikes, every GPS rate, whole zones held out and hikers the model never saw; vegetation and satellite embeddings don't transfer.

Next: Tobler with learned constants

✓ T1 dataset | ✓ T2 covariates | ✓ T3 four-model comparison | ✓ T4 slope curves | ✓ Paper draft 0

Weeks 10–11	Tobler's form with its constants learned from terrain, plus a variance head (PyTorch). Gate 2, 6 Nov: does it beat boosted trees?
Week 12	NAIP aerial imagery: a frozen pretrained encoder with a small head.
Week 13 to 11 Jan	Full paper draft, then IGARSS 2027 and its Student Paper Competition.

Question for the lab: which pretrained aerial-imagery encoder would you use for 1 m NAIP chips along trails?